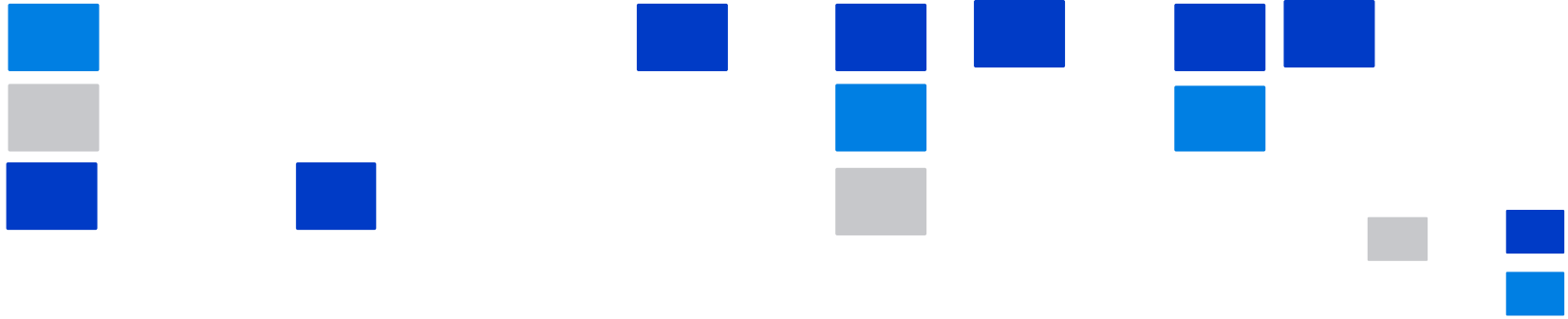




Sistemes Operatius

Tema 4: Crides al sistema - Threads • Curs 2026-27



Threads

- Generen fils d'execució.
- Comparteixen molts recursos (és àgil però alhora és un risc).
- Creació de threads: **`pthread_create()`**
- Esperar finalització: **`pthread_join()`** (equivalent a `wait()`).
- Si el fil principal mor, tots els threads es destrueixen.
- Compilar amb **`-lpthread`**.

Exemple 1

```
void *printf_xs(void *unused) {
    while (1) {
        write(STDERR_FILENO, "x", 1);
    }
    return NULL;
}

int main() {
    pthread_t thread_id;
    int status;

    status = pthread_create(&thread_id, NULL, printf_xs, NULL);
    if (status != 0) {
        exit(EXIT_FAILURE);
    }

    while (1) {
        write(STDERR_FILENO, "o", 1);
    }

    return EXIT_SUCCESS;
}
```

Exemple 2

```
void *threadFunc(void *arg) {
    char *s = (char *)arg;
    write(STDOUT_FILENO, s, strlen(s));
    return (void *)strlen(s);
}

int main(void) {
    pthread_t t1;
    void *res;
    char buffer[128];

    pthread_create(&t1, NULL, threadFunc, "Hello world\n");

    write(STDOUT_FILENO, "Message from main()\n", strlen("Message from main()\n"));

    pthread_join(t1, &res);

    snprintf(buffer, sizeof(buffer), "Thread returned %ld\n", (long)res);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    exit(EXIT_SUCCESS);
}
```

Exemple 2 (versió 2)

```
int main() {
    pthread_t t1;
    void *res;
    int status;
    char buffer[128];

    status = pthread_create(&t1, NULL, threadFunc, "Hello world\n");
    if (status != 0) {
        snprintf(buffer, sizeof(buffer), "Error create: %s\n", strerror(status));
        write(STDERR_FILENO, buffer, strlen(buffer));
        exit(EXIT_FAILURE);
    }

    write(STDOUT_FILENO, "Message from main()\n", strlen("Message from main()\n"));

    status = pthread_join(t1, &res);
    if (status != 0) {
        snprintf(buffer, sizeof(buffer), "Error join: %s\n", strerror(status));
        write(STDERR_FILENO, buffer, strlen(buffer));
        exit(EXIT_FAILURE);
    }

    snprintf(buffer, sizeof(buffer), "Thread returned %ld\n", (long)res);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    exit(EXIT_SUCCESS);
}
```



Amb retorn de les
funcions!!

Exemple 3: paràmetres

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

void *threadFunc(void *arg) {
    int *number = (int *)arg;

    *number = *number + 5;

    return (void *)number;
}
```

Exemple 3: paràmetres

```
int main() {  
    pthread_t t1;  
    void *res;  
    int x;  
    char buffer[128];  
  
    x = 7;  
  
    snprintf(buffer, sizeof(buffer), "Value of x=%d\n", x);  
    write(STDOUT_FILENO, buffer, strlen(buffer));  
  
    pthread_create(&t1, NULL, threadFunc, &x);  
  
    pthread_join(t1, &res);  
  
    snprintf(buffer, sizeof(buffer), "Thread returns %LX\n", (long)res);  
    write(STDOUT_FILENO, buffer, strlen(buffer));  
    snprintf(buffer, sizeof(buffer), "Value of x=%d\n", x);  
    write(STDOUT_FILENO, buffer, strlen(buffer));  
  
    exit(EXIT_SUCCESS);  
}
```

Exemple 3: paràmetres (versió 2)

```
int main() {
    pthread_t t1;
    void *res;
    int status, x;
    char buffer[128];

    x = 7;

    snprintf(buffer, sizeof(buffer), "Value of x=%d\n", x);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    status = pthread_create(&t1, NULL, threadFunc, &x);
    if (status != 0) {
        snprintf(buffer, sizeof(buffer), "Error create: %s\n", strerror(status));
        write(STDERR_FILENO, buffer, strlen(buffer));
        exit(EXIT_FAILURE);
    }

    status = pthread_join(t1, &res);
    if (status != 0) {
        snprintf(buffer, sizeof(buffer), "Error join: %s\n", strerror(status));
        write(STDERR_FILENO, buffer, strlen(buffer));
        exit(EXIT_FAILURE);
    }

    snprintf(buffer, sizeof(buffer), "Thread returns %p\n", res);
    write(STDOUT_FILENO, buffer, strlen(buffer));
    snprintf(buffer, sizeof(buffer), "Value of x=%d\n", x);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    exit(EXIT_SUCCESS);
}
```



Amb retorn de les
funcions!!

Exemple 4: threads i variables

```
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
```

```
int y;
```

```
void *threadFunc(void *arg) {
    int *number = (int *)arg;
    char buffer[64];

    *number = *number + 5;
    y = y + 1;

    snprintf(buffer, sizeof(buffer), "From thread y=%d\n", y);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    return arg;
}
```

Exemple 4: threads i variables

```
int main() {
    pthread_t t1;
    void *res;
    int x;
    char buffer[128];

    x = 7; y = 5;

    snprintf(buffer, sizeof(buffer), "[main] Initial value x=%d, y=%d\n", x, y);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    pthread_create(&t1, NULL, threadFunc, &x);

    y = y + 2;

    snprintf(buffer, sizeof(buffer), "[main] Middle value y=%d\n", y);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    pthread_join(t1, &res);

    snprintf(buffer, sizeof(buffer), "Thread returned %lX\n", (long)res);
    write(STDOUT_FILENO, buffer, strlen(buffer));
    snprintf(buffer, sizeof(buffer), "[main] Final value x=%d, y=%d\n", x, y);
    write(STDOUT_FILENO, buffer, strlen(buffer));

    exit(EXIT_SUCCESS);
}
```

Exemple 4: threads i variables (versió 2)

```
int main() {  
    pthread_t t1;  
    void *res;  
    int status, x;  
    char buffer[128];
```

```
    x = 7; y = 5;
```

```
    snprintf(buffer, sizeof(buffer), "[main] Initial value x=%d, y=%d\n", x, y);  
    write(STDOUT_FILENO, buffer, strlen(buffer));
```

```
    status = pthread_create(&t1, NULL, threadFunc, &x);  
    if (status != 0) {  
        write(STDERR_FILENO, "Error create\n", 13);  
        exit(EXIT_FAILURE);  
    }
```



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Exemple 4: threads i variables (versió 2)

```
y = y + 2;
```

```
snprintf(buffer, sizeof(buffer), "[main] Middle value y=%d\n", y);  
write(STDOUT_FILENO, buffer, strlen(buffer));
```

```
status = pthread_join(t1, &res);  
if (status != 0) {  
    write(STDERR_FILENO, "Error join\n", 11);  
    exit(EXIT_FAILURE);  
}
```

```
snprintf(buffer, sizeof(buffer), "Thread returned %lX\n", (long)res);  
write(STDOUT_FILENO, buffer, strlen(buffer));  
snprintf(buffer, sizeof(buffer), "[main] Final value x=%d, y=%d\n", x, y);  
write(STDOUT_FILENO, buffer, strlen(buffer));
```

```
exit(EXIT_SUCCESS);  
}
```



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